

NETFLIX — HIGH LEVEL SYSTEM DESIGN

Functional requirements • Non-functional requirements • Back-of-the-envelope estimation • Components • Flow & connections • ER/data relationships

FUNCTIONAL REQUIREMENTS	NON-FUNCTIONAL REQUIREMENTS	BACK-OF-THE-ENVELOPE ESTIMATION
• Sign up / Login / Profiles • Browse & search catalog • Personalized recommendations • Play / pause / resume video • Adaptive bitrate streaming • Watchlist & viewing history • Ratings / interactions • Subscriptions & payments • Multi-device support • Admin/content management	• High availability & fault tolerance • Massive scalability • Low startup latency • Global performance • Durability & consistency where needed • Security & privacy • Observability / monitoring • Disaster recovery • Cost efficiency • Millions of concurrent streams	Illustrative assumptions: • 300M subscribers / accounts • 100M daily active viewers • 20% concurrent at peak → 20M streams • Average bitrate ≈ 5 Mbps Peak video egress: 20M × 5 Mbps = 100 Tbps ≈ 12.5 TB/s 1 hr/day raw transfer: ≈ 225 PB/day 100M × 20 events/day = 2B events/day ≈ 23K events/s average; peak 100K+ events/s. <i>Assumptions, not Netflix production figures.</i>

ARCHITECTURE — COMPONENTS & CONNECTIONS

CLIENTS Web • Mobile • Smart TV • Console	DNS / GLOBAL TRAFFIC MANAGER	API GATEWAY / LOAD BALANCER Routing • Auth • Rate limits	MICROSERVICES Auth/User/Profile Catalog/Metadata Search Recommendation Playback/Session Subscription/Payment Watchlist/History
CONTENT PIPELINE Studios / Upload → Ingest → Transcode / Package → Object Storage	EVENT BUS / STREAM PROCESSING Kafka-like event stream • consumers • analytics	DATA LAYER User/Profile DB • Catalog DB • Subscription DB Cache • Search Index • Feature/Model Store • Data Lake/Warehouse	OBSERVABILITY & PLATFORM Metrics • Logs • Traces • Alerts • Autoscaling • Secrets
GLOBAL CDN / OPEN CONNECT STYLE EDGE CACHE Video segments • manifests • TLS • geo-distributed	OBJECT STORAGE Master / encoded video assets	RECOMMENDATION / ML PLATFORM Offline training → feature store → online ranking	SECURITY & RELIABILITY Encryption • IAM • retries • circuit breakers • disaster recovery

REQUEST / PLAYBACK FLOW

1. Client logs in / requests a title → API Gateway.	2. Catalog + Recommendation services return metadata / ranked titles.	3. Client selects Play → Playback service authorizes a session.	4. Manifest URL points to the nearest CDN edge.	5. Client downloads video chunks adaptively from CDN.	6. Playback events → Event Bus → Analytics + Recommendation pipeline.	7. Progress / history is persisted asynchronously; critical account/payment writes are durable.
---	---	---	---	---	---	---

SIMPLIFIED ER / DATA RELATIONSHIPS

USER user_id (PK) email plan_id (FK)	PROFILE profile_id (PK) user_id (FK) name	TITLE title_id (PK) name genre	WATCH_HISTORY history_id (PK) profile_id (FK) title_id (FK) progress	SUBSCRIPTION plan_id (PK) name status
--	---	--	---	---

Relationships: USER 1→N PROFILE • PROFILE 1→N WATCH_HISTORY • TITLE 1→N WATCH_HISTORY • SUBSCRIPTION 1→N USER